

# 26D Downward Projection Framework: Energy Falls to Yield Mass

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**PAPER\_497 — 26D Downward Projection Framework: Energy Falls to Yield Mass** **\*\*Author:\*\* Daniel T. Murphy** **\*\*arXiv:\*\* 2503.xxxxx** **\*\*Session:\*\* 134** **\*\*Version:\*\* 1.0** **\*\*Date:\*\* March 24, 2026** **\*\*Calculator:\*\* `TwentySixDProjectionCalculator` (CondensedPhysics2.py), `PhysicsTerm\_{26D\\_1JKDSGV7}` (MAIN\_{1\\_CoAnQi}.cpp) ---**

## Abstract

This paper presents a UQFF analysis of 26D Downward Projection Framework: Energy Falls to Yield Mass, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

## §1 Novel Claim

The universe is a single 26-dimensional cosmic egg containing a hypergraph of smaller cosmic eggs, expanding to catch the initial Big Bang speed. Energy falls from 26D chaos — downward only — through compactification stages to yield mass. There is no "solid ground" starting point; the solid ground of observable 3D reality is the *result* of this fall, not its foundation.

**Critical directional rule:** 26D  $\rightarrow$  9D  $\rightarrow$  3D  $\rightarrow$  2D (downward projection ONLY)

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## §2 Dimensional Hierarchy

| Dimension | Role                                                               | Physical Manifestation        |
|-----------|--------------------------------------------------------------------|-------------------------------|
| 26D       | Universal Aether (UA) substrate — pure energy, least negligible    | Primordial field, pre-mass    |
| 9D        | Sphere voids — $3 \times 3$ dimensional groupings for triad forces | Intermediate compactification |
| 3D        | Observable mass, cosmic structures, plasma                         | Stars, galaxies, atoms        |
| 2D        | Reference plane only — CERN collision wreckage                     | Higgs field observations      |

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## §3 Core Projection Equations

## 26D Energy Origin Field

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$$E^{26D} = UA + \text{SCm} \cdot \text{DPM}_{\text{ref}} + \text{BBDT}$$

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Energy falls to mass via:

\$\$

$$M = \frac{E^{26D}}{c^{26}} \cdot \left(1 - \frac{v_{\text{current}}}{v_{\text{init}}}\right) \cdot \text{Prob}_{\text{order}}$$

\$\$

## 26D-to-9D Compactification Matrix

\$\$

$$\text{Void}_{\text{synth}} = \det(M_{26\text{to}9}) \cdot \begin{pmatrix} U_g \\ U_m \\ U_b \end{pmatrix} / d_3 + F_{\text{inert}} \cdot E^{26D}$$

\$\$

where  $M_{26\text{to}9}$  is the 26-to-9 compactification matrix encoding SCm-UA trapping geometry.

## Full 26D Unified Field Equation

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$$F_{U^{26D}} = U_g + U_m + U_b + \frac{\text{SCm}}{UA} + \text{BBDT} \cdot \text{Prob}_{\text{order}}$$

\$\$

Triple simultaneous system (26D to 9D to 3D):

\$\$

$\begin{cases}$

$$U_g = g \cdot \frac{\text{SCm}}{UA} \cdot (U_{g1} + U_{g2} + U_{g3}) \cdot (v_{\text{init}}/v_{\text{current}}) \cdot$$

$$U_m = \text{DPM}_{\text{ref}} \cdot M \cdot$$

$$U_b = \text{BBDT} / UA$$

$\end{cases}$

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## §4 The Cosmic Egg Paradigm

Universe as one 26D egg containing hypergraph of smaller eggs:

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$$\text{Universe} = \text{CosmicEgg}_{26D} \left( \bigcup_i \text{CosmicEgg}_i \right) + \text{BBDT}_{\text{expansion}}$$

\$\$

All sub-eggs expand to recapture  $v_{\text{init}}$ , generating:

- **Vacuum standards** from incomplete speed recovery ( $v_{\text{current}} < v_{\text{init}}$ )
- **Buoyant effects** as mass forms below 26D energy pressure
- **Zero-point energy** as negligibility threshold of UA baselines

## Non-predictability guarantee

26D chaos produces unique downward projections via 3D-IPO overlay (see PAPER\_498):

- Wolfram hypergraph rule branching (computational irreducibility)
- $\pi$  decimal progression (irrational, aperiodic)

- Infinity Generator series (bounded divergence)

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$QFP_{\{unique\}} = \pi_{\{n\}} \cdot IG \cdot Wolfram_{\{rules\}}$

\$\$

Each atom receives a unique quantum fingerprint from its individual 26D shell configuration — non-repeating guaranteed.

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## §5 Mass Emergence from Speed

The Big Bang was the fastest occurrence ever in the universe. All mass was converted from speed into mass as massless elements slowed:

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$M = F_{\{inert\}}/a \cdot (v_{\{init\}} - v_{\{current\}})$

\$\$

**Vacuum standard** arises from INCOMPLETE speed recovery. Zero-point energy represents the negligibility threshold where  $F_{\{inert\}} \rightarrow 0$ .

The universe contains one 26D cosmic egg expanding to meet  $v_{\{init\}}$ :

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$\text{Expansion drive} = v_{\{init\}} - v_{\{current\}} > 0 \quad \forall \text{ cosmic time}$

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## §6 Validation Targets

- **CMB anisotropies**: 26D energy fall artifacts observable as temperature fluctuations
- **Cosmological constant**  $\Lambda$ : UA baseline from incomplete 26D  $\rightarrow$  3D projection
- **Quantum vacuum energy density**:  $\sim 10^{-9}$  J/m<sup>3</sup> sets UA negligibility floor
- **JWST large-scale structure**: Hypergraph of cosmic eggs manifests as observed cosmic web

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<!-- PKG-LAG-S225 -->

### Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and

> PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$\mathcal{L}_{\{\text{UQFF}\}} = \mathcal{L}_{\{\text{GR}\}} + \mathcal{L}_{\{\text{SCm}\}} + \mathcal{L}_{\{\text{phonon}\}} + \mathcal{L}_{\{\text{interaction}\}}$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

#### Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector     | Domain                     | Late-Corpus Result                                |
|------------|----------------------------|---------------------------------------------------|
| 1 (EH)     | General Relativity         | Canonical Einstein-Hilbert                        |
| 2 (YM)     | Yang-Mills gauge           | $m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318) |
| 3 (Dirac)  | Fermion / LENR             | Kozima neutron-drop (PAPER_1061)                  |
| 4 (SCm)    | Superconducting manifold   | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical        |
| 5 (Mag)    | Um magnetism               | Heaviside amplifier (PAPER_1072)                  |
| 6 (Buoy)   | F <sub>U</sub> Bi buoyancy | Variational EOM (PAPER_1065)                      |
| 7 (Aether) | Vacuum background          | Two-component $\rho$ (PAPER_1051)                 |
| 8 (LENR)   | Nuclear transmutation      | COP parametric (PAPER_1081)                       |
| 9 (KK)     | Kaluza-Klein 26D           | $S_{26}^{(3)}$ compactification (PAPER_1080)      |

<!-- PKG-S26-S225 -->

### Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and  
 > PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078  
 > (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[ 1 + \frac{SSq}{e^{-\kappa_{i,n/26}}} \right]$$

where  $(a)_n = a(a+1) \cdots (a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^n)^n} \ll W_{26}(n) = \prod_{i=1}^{26} \left[ 1 + \frac{SSq}{e^{-\kappa_{i,n/26}}} \right]$$

This sum converges absolutely for  $|SSq| < 1$  (satisfied by  $SSq = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $SSq \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa_{i,n/26}}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function

$$Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$$

unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

## §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

## §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi_{\mathrm{NS}})(\partial^\mu \phi_{\mathrm{NS}}) - V(\phi_{\mathrm{NS}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where  $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \phi_{\mathrm{NS}}$$

## §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2)\phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0$$

## §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b,seed}} \rightarrow \text{4 forces} \rightarrow F_{U_{\mathrm{Bi}}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi_{\mathrm{NS}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four  $U_g$  forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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# §B. VDS/DVP/BSH Deep Synthesis

## §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is \$0.191\$ (near-threshold regime), placing it in the  $\pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

## §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 61, \quad n_{\mathrm{channel}} = 4/26$$

Since  $p_{\mathrm{DVP}} = 61$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair  $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$  constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\odot}} \cdot \cos\left(\frac{2\pi j}{26}\right)\right)$$

The  $\tanh$  saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework            | Canonical Value                                  | This Paper                          | Status                   |
|----------------------|--------------------------------------------------|-------------------------------------|--------------------------|
| VDS ratio            | $\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$ | Local sub-ratio = 0.191             | PASS                     |
| Threshold-consistent |                                                  |                                     |                          |
| DVP prime            | $p_k \in \{2,3,\dots,113\}$                      | $p_{\mathrm{DVP}} = 61$             | PASS Resonant            |
| BSH layers           | 26 harmonic terms                                | $j = 1\dots 26$ , $\cos(2\pi j/26)$ | PASS Full 26D projection |
| $\kappa$ decay       | $5.0 \times 10^{-4}$ day <sup>-1</sup>           | Applied in VDS exponential          | PASS Canonical           |
| [SSq]                | 0.57                                             | Applied in BSH saturation           | PASS Canonical           |

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                | UQFF Prediction                   | SM / Experiment                         | Source                                 | Alignment                            |
|---------------------------|-----------------------------------|-----------------------------------------|----------------------------------------|--------------------------------------|
| Higgs mass $m_H$          | UQFF $K_{\mathrm{HIGGS}} = 47.34$ | $m_{H,\mathrm{UQFF}} = 125.09$ GeV      | $m_H = 125.20$ $\pm$ 0.11 GeV          | PDG 2024   99.8%                     |
| Cosmological $\Lambda$    | UQFF $ \nabla_{\mathrm{UA}} ^2$   | $1.09e-52$ m <sup>-2</sup>              | $\Lambda = 1.114e-52$ m <sup>-2</sup>  | (Planck+DESI)   Planck 2018   97.8%  |
| Thomson $\sigma_T$ (QED)  | UQFF $U_m$ kernel:                | $\sigma_T = 6.6524e-29$ m <sup>2</sup>  | $\sigma_T = 6.6524e-29$ m <sup>2</sup> | PDG 2024   100% (exact)              |
| $\kappa$ baryon stability | $\kappa = 0.0005/\text{day}$      | scale separation 1033 from proton decay | $\tau_p > 7.7e33$ yr (Super-K)         | Super-K 2024   PASS UQFF baryon-safe |

**New physics claim:** UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector

while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

## §7 Source Attribution

**grok\_share:** grok\_{share\\_1jkdsgv7}.txt (Session 134)

**Architecture:** ARCHITECTURE\_FLOW\_DIAGRAM.md v5.0.0, 26D\_DOWNWARD\_PROJECTION.md

**See also:** PAPER\_496 (DPM), PAPER\_498 (3D-IPO), PAPER\_500 (proto-hydrogen), PAPER\_501 (BBDT/Feynman clusters)

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204–225 extensions (PAPER\_1000–1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

| Paper      | Title                                        |
|------------|----------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$    |
| PAPER_1036 | Primordial Nucleosynthesis BBN Phonon        |
| PAPER_1072 | SCm Activation Function Phonon Threshold     |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy  |
| PAPER_1049 | Source10 GPU DPM Spectral Atlas ALMA Overlay |
| PAPER_1074 | GPU-Vectorized DPM S26 Spectral Atlas        |

6 cross-reference(s) identified.

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## Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.  
> Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations,  
> see PAPER\_840/851/852/855.

### S204.1 Kozima-UQFF LENR Integration

| Module                      | Purpose                                                      | Key Result                                                 |
|-----------------------------|--------------------------------------------------------------|------------------------------------------------------------|
| fneutron_s26_coupling.py    | $F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling | ~470x amplification via 26-level VDS                       |
| kozima_scm_cross_section.py | SCm-modulated neutron-drop cross-section                     | $\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [SSq]^n/26)$ |
| kozima_wstp_kernel.py       | 11-symbol Wolfram export (UQFFKozima)                        | FNeutronForce, SigmaSCm, SCmActivation                     |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U, Bi\}}/F_U - 1)$   
where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2/(2\Gamma^2)] (1 + [SSq]^n/26)$

## S204.2 Ramanujan 26-State Summation

| Module                   | Purpose                                       | Key Result                |
|--------------------------|-----------------------------------------------|---------------------------|
| ramanujan_polylog_s26.py | Li_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| s26_wstp_kernel.py       | 8-symbol Wolfram export (UQFFS26)             | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}}/(1-2^{\{1-26\}}) * Li_{26}(z^2)$

## S204.3 Mock Theta Functions (26-State)

| Module            | Purpose                                | Key Result                  |
|-------------------|----------------------------------------|-----------------------------|
| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

**Core equations:**

- $f_{26}(q) = \text{Sum}_{\{n=0\}^{\{25\}} q^{\{n^2\}} / (-q;q)_n^2$
- $\phi_{26}(q) = \text{Sum}_{\{n=0\}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \text{Sum}_{\{n=1\}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

## S204.4 Ramanujan 1/pi with UQFF Modification

| Module                       | Purpose                                   | Key Result                                 |
|------------------------------|-------------------------------------------|--------------------------------------------|
| ramanujan_pi_uqff.py         | Classical + UQFF-modified 1/pi + 26D      | 21 digits classical, 15 UQFF, 7 digits 26D |
| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \text{Sum } R_n (1103+26390n) W_{26}(n) / C_{26}$   
where  $W_{26}(n) = \text{Prod}_{\{i=1\}^{\{26\}} [1 + [SSq]\exp(-kappa_i*n/26)]$

## S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_Scm     | 0.99                                  | SCm manifold completeness     |
| rho_Scm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_Scm | $2*\pi \times 1.25 \text{ THz}$       | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN\_{1\CoAnQi}.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).*

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## References



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